

Data Analysis Re-visit

20 Fundamental Questions for NumPy, Pandas, Matplotlib & Seaborn

Section 1: NumPy Basics

- 1. Array Creation:** Create a 1D NumPy array with integers from 10 to 16. What is its shape?
- 2. Mean & Sum:** Given the array `np.array([120, 150, 90, 200, 250, 180, 210])`, calculate the mean and total sum.
- 3. Array Filtering:** Write code to extract all numbers greater than 150 from `np.array([100, 160, 140, 220, 180])`.
- 4. Reshaping:** Reshape a 1D array of 6 elements into a 2D array with 2 rows and 3 columns.
- 5. Element-wise Math:** Multiply all elements of `np.array([10, 20, 30])` by 1.1 (a 10% increase).

Section 2: Pandas Fundamentals

- 6. DataFrame Creation:** Create a DataFrame from a dictionary with two keys: `'Day' ([ 'Mon', 'Tue', 'Wed' ])` and `'Sales' ([100, 150, 200])`.
- 7. Filtering Rows:** Filter a DataFrame `df` to keep only rows where the `'Sales'` column is strictly greater than 120.
- 8. Selecting Columns:** Write the code to select only the `'Day'` and `'Sales'` columns from a larger DataFrame `df`.
- 9. Adding Columns:** Add a new column named `'Tax'` to a DataFrame `df`, calculated as 10% (`0.10`) of the `'Sales'` column.
- 10. Summary Statistics:** What single Pandas method provides quick summary statistics (mean, std, min, max) for numeric columns?
- 11. Handling Missing Data:** How do you drop all rows containing missing (`NaN`) values in a Pandas DataFrame?
- 12. Sorting Data:** Sort a DataFrame `df` by the `'Sales'` column in descending order.

Section 3: Matplotlib Essentials

13. Line Plot: Write the basic Matplotlib code to plot `days = [ 'Mon', 'Tue', 'Wed' ]` on the x-axis and `sales = [100, 150, 200]` on the y-axis.

14. Titles & Labels: Add the title `"Weekly Performance"` and label the y-axis `"Revenue ($)"` to a Matplotlib plot.

15. Displaying Plots: What function call is required at the end of your script to display a Matplotlib figure?

16. Scatter Plot: Which Matplotlib function creates a scatter plot instead of a line plot?

Section 4: Seaborn & Data Visualization

17. Bar Plot: Write a Seaborn command (`sns.barplot`) to create a bar plot using a DataFrame `df` with `'Day'` on the x-axis and `'Sales'` on the y-axis.

18. Plot Themes: How do you set the aesthetic theme in Seaborn to `'darkgrid'` ?

19. Histogram / Distribution: Which Seaborn function generates a distribution plot (histogram with optional KDE curve) for a single numeric column?

20. Box Plot: Write the Seaborn syntax to draw a box plot showing the distribution of `'Sales'` grouped by `'Category'` .